tion for monitoring the health condition of spindle and machine tool axes.

Object Management Group is helping with IIoT standardisation across the globe. Data Distribution Service (DDS), Systems Modeling Language (SysML) and Interaction Flow Modeling Language (IFML) are contributing to improve the productivity, cut costs and streamline processes of the IIoT.

IIoT no longer a hype

Earlier, the IIoT was an application that required lots of capital investment. So only up to 15 per cent of industrial assets was converted into digitised data. Now, as enterprises aim to lower sensor costs and change the route of sensor application, the IIoT is said to convert assets up to 35 per cent.

Initiatives have been taken in European countries like Germany and France to promote implementation of the IIoT on a big scale. Countries like USA, India and China are also deploying smart manufacturing technologies to increase their national GDPs. Schneider Electric has recently expanded its automation portfolio in India with the launch of IoT-enabled smart pumping solutions, which asserts that the IIoT is no longer a hype.

IIoT and smart applications

From smart dust to drones, futuristic farming to energy networks, motor control to machine-to-machine, IIoT applications are creating a web across the globe.

Motors are a generic element of the manufacturing industry. So their efficiency, reliability and power consumption need to be taken care of. FPGA ICs exhibit long-term reliability and long lifecycles, supporting the working of motors for long hours.

Similarly, there are devices that support the powerful Ethernet-based protocol for real-time communications.

5G is the latest trend, which has left 2G, 3G and 4G behind. This technology is evoking the interest of key industry players due to its higher bitrate, lower latency and extended bandwidth. For instance, smart cars and intelligent mobility systems largely depend upon 5G and the IoT. Smart cars provide sophisticated features such as infotainment, communications, diagnostics and driver assistance. Fully automated vehicles are still far off, yet some manufacturers are providing smart features like parking assistance and crash avoidance.

According to research reports, by year 2020, connected cars will account for up to 75 per cent of the worldwide shipments, which is quite large compared to the present scenario. And embedded systems will account for 40 per cent of the material cost of these